

Talgat Zharkynbay

“Module 4 Homework”

Load packages

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 3.5.3
```

```
library(stargazer)
```

```
## Warning: package 'stargazer' was built under R version 3.5.2
```

```
require(aod)
```

```
## Warning: package 'aod' was built under R version 3.5.3
```

```
require(car)
```

```
## Warning: package 'car' was built under R version 3.5.3
```

```
## Warning: package 'carData' was built under R version 3.5.2
```

```
library(Hmisc)
```

```
## Warning: package 'Hmisc' was built under R version 3.5.3
```

```
## Warning: package 'lattice' was built under R version 3.5.3
```

```
## Warning: package 'survival' was built under R version 3.5.3
```

```
## Warning: package 'Formula' was built under R version 3.5.2
```

```
## Warning: package 'ggplot2' was built under R version 3.5.3
```

```
library(lmtest)
```

```
## Warning: package 'lmtest' was built under R version 3.5.3
```

```
## Warning: package 'zoo' was built under R version 3.5.3
```

```
library(sandwich)
```

```
## Warning: package 'sandwich' was built under R version 3.5.3
```

Q1

```
#i)
FERTIL=read.table(file="FERTIL.txt", sep="", header=FALSE, as.is = TRUE, na.strings = c("NA",
".", ""))
colnames(FERTIL)=c("GFR", "PE", "YEAR")
FERTIL$WW2=ifelse(FERTIL$YEAR>=1941 & FERTIL$YEAR<=1945, 1, 0)
FERTIL$PILL=ifelse(FERTIL$YEAR>=1963,1,0)
FERTIL=FERTIL[order(FERTIL$YEAR),]
FERTIL$PE_LAG1=Lag(FERTIL$PE,1)
FERTIL$PE_LAG2=Lag(FERTIL$PE,2)
#ii)
Fit1=lm(GFR~PE+PE_LAG1+PE_LAG2, data = FERTIL)
SummaryFit1=summary(Fit1)
SummaryFit1
```

```
##
## Call:
## lm(formula = GFR ~ PE + PE_LAG1 + PE_LAG2, data = FERTIL)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.146 -16.062  -0.127   17.562   31.842
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  93.15791    4.49965   20.703  <2e-16 ***
## PE           -0.01584    0.14026   -0.113   0.910
## PE_LAG1      -0.02134    0.21523   -0.099   0.921
## PE_LAG2       0.05390    0.13811    0.390   0.698
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.78 on 66 degrees of freedom
## (2 observations deleted due to missingness)
## Multiple R-squared:  0.006135, Adjusted R-squared:  -0.03904
## F-statistic: 0.1358 on 3 and 66 DF, p-value: 0.9383
```

#The impact propensity multiplier of PE is estimate of the b1 coefficient and is equal to -0.0158. With a t-value of -0.11 and p-value of 0.91, it is insignificant at the 5% level of the test.

#The estimated effect of a one-dollar change in PE at time t on GFR at time t+1 is equal to the sum of the estimates of the b1 and b2 coefficients and is equal to -0.0158-0.0213=-0.0371. We use F-test to check if the effect is significantly different from zero:

```
linearHypothesis(Fit1, c("PE+PE_LAG1=0"), test = "F")
```

```
## Linear hypothesis test
##
## Hypothesis:
## PE + PE_LAG1 = 0
##
## Model 1: restricted model
## Model 2: GFR ~ PE + PE_LAG1 + PE_LAG2
##
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1      67 25861
## 2      66 25833   1    28.218 0.0721 0.7891
```

#With a high p-value of 0.79, we can not reject the hypothesis that effect is not significantly different from zero.

#The LRP is equal to the sum of the estimates of the coefficients of $b_1+b_2+b_3$, or $-0.0158-0.0213+0.0539=0.0168$. Hence, we can test the significance with the following F-test:
`linearHypothesis(Fit1, c("PE+PE_LAG1+PE_LAG2=0"), test = "F" )`

```
## Linear hypothesis test
##
## Hypothesis:
## PE + PE_LAG1 + PE_LAG2 = 0
##
## Model 1: restricted model
## Model 2: GFR ~ PE + PE_LAG1 + PE_LAG2
##
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1      67 25912
## 2      66 25833   1    78.627 0.2009 0.6555
```

#With a high p-value of 0.66, we can not reject the hypothesis that LRP is not significantly different from zero.

```
#iii)
FERTIL = FERTIL %>%
  mutate(X2 = PE_LAG1-PE, X3=PE_LAG2-PE)
Fit1_LRP_CI=lm(GFR~PE+X2+X3, data = FERTIL)
confint(Fit1_LRP_CI, "PE", level = 0.95, alternative="two.sided")
```

```
##           2.5 %      97.5 %
## PE -0.05775974 0.09119878
```

#This is the 95% CI for LRP, we can see that it contains zero value, which is consistent with the hypothesis we tested above.

```
#iv)
#First, we add the time trend:
FERTIL$t=1:length(FERTIL$YEAR)
FERTIL = FERTIL %>%
  mutate(t_sq = t^2)
#Next, we estimate our 3 Models:
Fit2_1=lm(GFR~PE+WW2+PILL, data = FERTIL)
Fit2_2=lm(GFR~PE+WW2+PILL+t, data = FERTIL)
Fit2_3=lm(GFR~PE+WW2+PILL+t+t_sq, data = FERTIL)
```

```
stargazer(Fit2_1, Fit2_2, Fit2_3, title="Regression Results", align=TRUE,type="html")
```

Regression Results

	Dependent variable:		
	GFR		
	(1)	(2)	(3)
PE	0.083*** (0.030)	0.279*** (0.040)	0.348*** (0.040)
WW2	-24.238*** (7.458)	-35.592*** (6.297)	-35.880*** (5.708)
PILL	-31.594*** (4.081)	0.997 (6.262)	-10.120 (6.336)
t		-1.150*** (0.188)	-2.531*** (0.389)
t_sq			0.020*** (0.005)
Constant	98.682*** (3.208)	111.769*** (3.358)	124.092*** (4.361)
Observations	72	72	72
R ²	0.473	0.662	0.727
Adjusted R ²	0.450	0.642	0.706
Residual Std. Error	14.685 (df = 68)	11.849 (df = 67)	10.739 (df = 66)
F Statistic	20.378*** (df = 3; 68)	32.837*** (df = 4; 67)	35.094*** (df = 5; 66)
Note:	p<0.1; p<0.05 ; p<0.01		

#The first model that ignores time trends predicts that a 1\$ increase in the average real dollar value of the personal tax exemption immediately increases the number of children born to every 1,000 women by 0.083.

#The second model take into account the linear time trend, because GFR in the US displays a clear downward trend over the years. This has increased the estimate of the PE to 0.279, hence a 1\$ increase in the average real dollar value of the personal tax exemption immediately increases the number of children born to every 1,000 women by 0.279. Hence, after the detrending of the data, we learn that PE actually has larger effect on GFR.

#The third model uses quadratic time trend idea. Since b4 is negative and b5 is positive, we can infer that changes in GFR due to the passage of time has a U-shape. The coefficient of PE is still highly significant with a value of 0.348, which means that PE actually has even higher effect on GFR.

The third model seems to have the best explanation of the variation in GFR, since it includes the quadratic time trend of GFR. The coefficient estimate of t is equal to -2.531, and the coefficient estimate of t_sq is equal to 0.020. Hence, each year the GFR decreases by $-2.531 + 2 \cdot 0.020 = -2.49$. Model 1 and Model 2 underestimate the effect of PE on GFR, because data is not fully detrended.

The estimates of R² and adjusted R² for detrended models contain the upward bias and overestimate the power of the model. Hence, we need to properly find the measures of the R²:

```

Fit2_Detrend=lm(GFR~t, data=FERTIL)
GFR_resid=Fit2_Detrend$residuals
Fit2_Appropriate=lm(GFR_resid~PE+WW2+PILL+t, data=FERTIL)

Fit2_Detrend_t_sq=lm(GFR~t+t_sq, data=FERTIL)
GFR_resid_t_sq=Fit2_Detrend$residuals
Fit2_Appropriate_t_sq=lm(GFR_resid~PE+WW2+PILL+t+t_sq, data=FERTIL)

Regout=stargazer(Fit2_Appropriate, Fit2_Appropriate_t_sq,title="Appropriate Goodness-Of-Fit",
align=TRUE,type="html")

```

Appropriate Goodness-Of-Fit

	<i>Dependent variable:</i>	
	GFR_resid	
	(1)	(2)
PE	0.279*** (0.040)	0.348*** (0.040)
WW2	-35.592*** (6.297)	-35.880*** (5.708)
PILL	0.997 (6.262)	-10.120 (6.336)
t	-0.641*** (0.188)	-2.022*** (0.389)
t_sq		0.020*** (0.005)
Constant	-2.453 (3.358)	9.870** (4.361)
Observations	72	72
R ²	0.524	0.615
Adjusted R ²	0.496	0.586
Residual Std. Error	11.849 (df = 67)	10.739 (df = 66)
F Statistic	18.473*** (df = 4; 67)	21.104*** (df = 5; 66)

Note: $p < 0.1$; **$p < 0.05$** ; $p < 0.01$

#The results confirm our statement that Model 3 has the better explanation. It's appropriate goodness-of-fit measure is equal to 0.586, which is higher than linear trend Model 2 with adjusted R² of 0.496, and simple model of 0.450.

Q2

```

#i)
Traffic=read.csv(file="TRAFFIC2_R.csv", header=FALSE, as.is=TRUE,sep = ",", na.strings = c("N
A", ".", ""))

View(Traffic)
Traffic=Traffic[, c("V1","V2", "V3","V4", "V5", "V6", "V7", "V8", "V9", "V10" )]
colnames(Traffic)=c("YEAR", "MONTH", "TOTACC", "FATAACC", "INJACC", "PDOACC", "UNEM","SPDLAW",
"BELTLAW","WKENDS")
Traffic$BELTLAW_lag1=Lag(Traffic$BELTLAW, 1)
Traffic$SPDLAW_lag1=Lag(Traffic$SPDLAW, 1)
Traffic[ Traffic$BELTLAW-Traffic$BELTLAW_lag1==1, c(1,2)]

```

```
##      YEAR MONTH
## NA      NA      NA
## 61 1986      1
```

```
Traffic[ Traffic$SPDLAW-Traffic$SPDLAW_lag1==1, c(1,2)]
```

```
##      YEAR MONTH
## NA      NA      NA
## 77 1987      5
```

```
#The belt law took effect in 1986 and the speed law took effect in 1987.
#ii)
Traffic$t= 1:length(Traffic$YEAR)
Fit3=lm(log(TOTACC)~ t + factor(MONTH), data=Traffic)
summaryFit3=summary(Fit3)
summaryFit3
```

```
##
## Call:
## lm(formula = log(TOTACC) ~ t + factor(MONTH), data = Traffic)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.100216 -0.039778 -0.002915  0.038747  0.138015
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   10.4685664   0.0190028  550.895 < 2e-16 ***
## t              0.0027471   0.0001611   17.057 < 2e-16 ***
## factor(MONTH)2 -0.0426864   0.0244475   -1.746  0.084036 .
## factor(MONTH)3  0.0798245   0.0244491    3.265  0.001523 **
## factor(MONTH)4  0.0184850   0.0244517    0.756  0.451532
## factor(MONTH)5  0.0320983   0.0244554    1.313  0.192506
## factor(MONTH)6  0.0201919   0.0244602    0.825  0.411156
## factor(MONTH)7  0.0375825   0.0244660    1.536  0.127835
## factor(MONTH)8  0.0539828   0.0244729    2.206  0.029808 *
## factor(MONTH)9  0.0423611   0.0244809    1.730  0.086810 .
## factor(MONTH)10 0.0821134   0.0244899    3.353  0.001149 **
## factor(MONTH)11 0.0712786   0.0244999    2.909  0.004511 **
## factor(MONTH)12 0.0961571   0.0245110    3.923  0.000165 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05186 on 95 degrees of freedom
## Multiple R-squared:  0.7969, Adjusted R-squared:  0.7712
## F-statistic: 31.06 on 12 and 95 DF,  p-value: < 2.2e-16
```

```
bptest(Fit3, studentize = FALSE)
```

```
##
## Breusch-Pagan test
##
## data: Fit3
## BP = 7.2528, df = 12, p-value = 0.8405
```

#Firstly, we see that coefficient estimate on the time trend is highly significant and equal to 0.002747. Hence we have a Log-Level relationship, and each year the number of traffic accidents increases by 0.275% on average. The BP test shows a high p-value, hence we accept homoskedasticity. To test the effect of seasonality, we can do the F-test:

```
Traffic$FEB=ifelse(Traffic$MONTH=="2", 1,0)
Traffic$MAR=ifelse(Traffic$MONTH=="3", 1,0)
Traffic$APR=ifelse(Traffic$MONTH=="4", 1,0)
Traffic$MAY=ifelse(Traffic$MONTH=="5", 1,0)
Traffic$JUN=ifelse(Traffic$MONTH=="6", 1,0)
Traffic$JUL=ifelse(Traffic$MONTH=="7", 1,0)
Traffic$AUG=ifelse(Traffic$MONTH=="8", 1,0)
Traffic$SEP=ifelse(Traffic$MONTH=="9", 1,0)
Traffic$OCT=ifelse(Traffic$MONTH=="10", 1,0)
Traffic$NOV=ifelse(Traffic$MONTH=="11", 1,0)
Traffic$DEC=ifelse(Traffic$MONTH=="12", 1,0)

Fit3=lm(log(TOTACC)~ t + FEB+MAR+APR+MAY+JUN+JUL+AUG+SEP+OCT+
        NOV+DEC, data=Traffic)

linearHypothesis(Fit3, c("FEB=0", "MAR=0", "APR=0", "MAY=0", "JUN=0",
                        "JUL=0", "AUG=0", "SEP=0", "OCT=0", "NOV=0", "DEC=0"
                        ), test = "F")
```

```
## Linear hypothesis test
##
## Hypothesis:
## FEB = 0
## MAR = 0
## APR = 0
## MAY = 0
## JUN = 0
## JUL = 0
## AUG = 0
## SEP = 0
## OCT = 0
## NOV = 0
## DEC = 0
##
## Model 1: restricted model
## Model 2: log(TOTACC) ~ t + FEB + MAR + APR + MAY + JUN + JUL + AUG + SEP +
##          OCT + NOV + DEC
##
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     106 0.40786
## 2      95 0.25550 11   0.15236 5.1501 2.712e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#The p-value is very low, hence we can conclude that there is a seasonality in total accidents.

#iii)

```
Fit4=lm(log(TOTACC)~ t + factor(MONTH)+UNEM+SPDLAW+BELTLAW+WKENDS, data=Traffic)
summaryFit4=summary(Fit4)
summaryFit4
```

```
##
## Call:
## lm(formula = log(TOTACC) ~ t + factor(MONTH) + UNEM + SPDLAW +
##     BELTLAW + WKENDS, data = Traffic)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.084489 -0.023998 -0.002176  0.024586  0.089315
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   10.6398628   0.0630859  168.657 < 2e-16 ***
## t              0.0011011   0.0002579    4.270 4.79e-05 ***
## factor(MONTH)2 -0.0338346   0.0177683   -1.904 0.06004 .
## factor(MONTH)3  0.0769530   0.0167941    4.582 1.46e-05 ***
## factor(MONTH)4  0.0104562   0.0170469    0.613 0.54115
## factor(MONTH)5  0.0237075   0.0169389    1.400 0.16504
## factor(MONTH)6  0.0219334   0.0172149    1.274 0.20587
## factor(MONTH)7  0.0499292   0.0167036    2.989 0.00360 **
## factor(MONTH)8  0.0559523   0.0168173    3.327 0.00127 **
## factor(MONTH)9  0.0420694   0.0172819    2.434 0.01687 *
## factor(MONTH)10 0.0817170   0.0169554    4.820 5.73e-06 ***
## factor(MONTH)11 0.0768721   0.0172455    4.458 2.36e-05 ***
## factor(MONTH)12 0.0990862   0.0170705    5.805 9.32e-08 ***
## UNEM           -0.0212173   0.0033974   -6.245 1.33e-08 ***
## SPDLAW          -0.0537593   0.0126036   -4.265 4.87e-05 ***
## BELTLAW         0.0954526   0.0142350    6.705 1.65e-09 ***
## WKENDS          0.0033333   0.0037761    0.883 0.37970
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03524 on 91 degrees of freedom
## Multiple R-squared:  0.9101, Adjusted R-squared:  0.8943
## F-statistic: 57.61 on 16 and 91 DF, p-value: < 2.2e-16
```

```
bptest(Fit4, studentize = FALSE)
```

```
##
## Breusch-Pagan test
##
## data: Fit4
## BP = 16.532, df = 16, p-value = 0.4165
```


#We again accept homoskedasticity. With regards to UNEM estimates, the regression predicts that on average the number of traffic accidents decreases by 2.12% with a 1 unit increase in unemployment rate. This might make sense, since unemployed people might have less money to buy cars and hence they will have less car accidents. With the introduction of Speedlaw the number of traffic accidents decreases by 5.38% which is sort of expected. However, with the introduction of Beltlaw the number of traffic accidents increases by 9.55%, which is very unexpected.

#iv)

```
Traffic$PRCFAT=(Traffic$FATAACC/Traffic$TOTACC)*100  
mean(Traffic$PRCFAT)
```

```
## [1] 0.8856363
```

#The results suggest that 0.886% of total accidents are fatal. In other words, almost 1 out of 100 car accidents results in death of at least one passenger. The magnitude does not seem to be unusual.

#v)

```
Fit5=lm(PRCFAT~ t + factor(MONTH)+UNEM+SPDLAW+BELTLAW+WKENDS, data=Traffic)  
summaryFit5=summary(Fit5)  
summaryFit5
```

```
##
## Call:
## lm(formula = PRCFAT ~ t + factor(MONTH) + UNEM + SPDLAW + BELTLAW +
##     WKENDS, data = Traffic)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.132085 -0.032573  0.000621  0.038226  0.132080
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.030e+00  1.030e-01  10.003 2.52e-16 ***
## t             -2.235e-03  4.208e-04  -5.312 7.66e-07 ***
## factor(MONTH)2  8.608e-04  2.900e-02   0.030 0.976383
## factor(MONTH)3  9.228e-05  2.741e-02   0.003 0.997321
## factor(MONTH)4  5.822e-02  2.782e-02   2.093 0.039152 *
## factor(MONTH)5  7.164e-02  2.764e-02   2.592 0.011129 *
## factor(MONTH)6  1.013e-01  2.809e-02   3.604 0.000510 ***
## factor(MONTH)7  1.766e-01  2.726e-02   6.479 4.64e-09 ***
## factor(MONTH)8  1.926e-01  2.744e-02   7.018 3.91e-10 ***
## factor(MONTH)9  1.600e-01  2.820e-02   5.674 1.64e-07 ***
## factor(MONTH)10 1.010e-01  2.767e-02   3.651 0.000435 ***
## factor(MONTH)11 1.395e-02  2.814e-02   0.496 0.621345
## factor(MONTH)12 9.201e-03  2.786e-02   0.330 0.741959
## UNEM          -1.543e-02  5.544e-03  -2.782 0.006563 **
## SPDLAW         6.709e-02  2.057e-02   3.262 0.001560 **
## BELTLAW       -2.951e-02  2.323e-02  -1.270 0.207287
## WKENDS         6.259e-04  6.162e-03   0.102 0.919328
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05751 on 91 degrees of freedom
## Multiple R-squared:  0.7174, Adjusted R-squared:  0.6677
## F-statistic: 14.44 on 16 and 91 DF,  p-value: < 2.2e-16
```

#The belt law decreases the percentage of fatal accidents by 0.029%, but the estimate is statistically insignificant. The speed law, on the contrary increased the number of fatal accidents by 0.067% and is statistically significant, which is very counterintuitive. Hence, it seems that the introduction of the laws did not reach the desired effects.

```
#vi)
Traffic$PRCFAT_lag1=Lag(Traffic$PRCFAT, 1)
Traffic$UNEM_lag1=Lag(Traffic$UNEM, 1)
Fit6=lm(PRCFAT~PRCFAT_lag1, data=Traffic)
summaryFit6=summary(Fit6)
summaryFit6
```

```
##
## Call:
## lm(formula = PRCFAT ~ PRCFAT_lag1, data = Traffic)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.141962 -0.036649 -0.004246  0.038963  0.207079
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.25690     0.06146   4.18 6.06e-05 ***
## PRCFAT_lag1  0.70939     0.06893  10.29 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07106 on 105 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.5022, Adjusted R-squared:  0.4974
## F-statistic: 105.9 on 1 and 105 DF,  p-value: < 2.2e-16
```

```
bptest(Fit6, studentize = FALSE) # accept homoskedasticity
```

```
##
## Breusch-Pagan test
##
## data:  Fit6
## BP = 0.56756, df = 1, p-value = 0.4512
```

```
confint(Fit6, "PRCFAT_lag1", level = 0.95, alternative="two.sided")
```

```
##              2.5 %    97.5 %
## PRCFAT_lag1 0.5727166 0.8460656
```

#We are not worried about the unit root, since both the CI and point estimate indicate that r is less than 0.8, and the estimate is significant.

```
Fit7=lm(UNEM ~ UNEM_lag1, data=Traffic)
summaryFit7=summary(Fit7)
summaryFit7
```

```
##
## Call:
## lm(formula = UNEM ~ UNEM_lag1, data = Traffic)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.13187 -0.43997 -0.01477  0.30531  1.42201
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.27978     0.22893   1.222   0.224
## UNEM_lag1    0.95726     0.03078  31.103 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.565 on 105 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.9021, Adjusted R-squared:  0.9012
## F-statistic: 967.4 on 1 and 105 DF,  p-value: < 2.2e-16
```

```
bptest(Fit7, studentize = FALSE) # accept homoskedasticity
```

```
##
## Breusch-Pagan test
##
## data:  Fit7
## BP = 1.1109, df = 1, p-value = 0.2919
```

```
confint(Fit7, "UNEM_lag1", level = 0.95, alternative="two.sided")
```

```
##              2.5 %    97.5 %
## UNEM_lag1 0.8962318 1.018282
```

#Here, we are worried about the unit root, because the estimate is very close to 1 and significant, meaning that the variable follows a random walk, hence the usual inference procedures will not be valid.

#vii)

```
Traffic$PRCFAT_DIF=Traffic$PRCFAT-Traffic$PRCFAT_lag1
```

```
Traffic$UNEM_DIF=Traffic$UNEM-Traffic$UNEM_lag1
```

```
Fit8=lm(PRCFAT_DIF~UNEM_DIF+SPDLAW+BELTLAW+WKENDS+t+FEB+MAR+APR+MAY+JUN+JUL+AUG+SEP+OCT+NOV+DEC,data=Traffic)
```

```
SummaryFit8=summary(Fit8)
```

```
SummaryFit8
```

```
##
## Call:
## lm(formula = PRCFAT_DIF ~ UNEM_DIF + SPDLAW + BELTLAW + WKENDS +
##      t + FEB + MAR + APR + MAY + JUN + JUL + AUG + SEP + OCT +
##      NOV + DEC, data = Traffic)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.130088 -0.044411  0.002195  0.032828  0.166416
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.1268680   0.1048114  -1.210   0.22928
## UNEM_DIF      0.0125342   0.0161094   0.778   0.43857
## SPDLAW       -0.0071825   0.0237979  -0.302   0.76349
## BELTLAW       0.0008251   0.0265048   0.031   0.97523
## WKENDS        0.0068097   0.0072276   0.942   0.34862
## t             0.0001433   0.0004849   0.296   0.76823
## FEB           0.0346229   0.0370460   0.935   0.35250
## MAR           0.0419346   0.0389248   1.077   0.28422
## APR           0.0985703   0.0382989   2.574   0.01170 *
## MAY           0.0568102   0.0374416   1.517   0.13270
## JUN           0.0540340   0.0347738   1.554   0.12373
## JUL           0.0878394   0.0331103   2.653   0.00943 **
## AUG           0.0589255   0.0396687   1.485   0.14092
## SEP           0.0065431   0.0379741   0.172   0.86358
## OCT          -0.0323897   0.0352025  -0.920   0.35998
## NOV          -0.0591083   0.0354151  -1.669   0.09859 .
## DEC           0.0272794   0.0363246   0.751   0.45462
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06724 on 90 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.3436, Adjusted R-squared:  0.2269
## F-statistic: 2.945 on 16 and 90 DF,  p-value: 0.0006167
```

```
stargazer(Fit5, Fit8, title="Non-difference vs Difference", align=TRUE, type="html")
```

Non-difference vs Difference

	<i>Dependent variable:</i>	
	PRCFAT	PRCFAT_DIF
	(1)	(2)
UNEM_DIF		0.013 (0.016)
t	-0.002*** (0.0004)	0.0001 (0.0005)
factor(MONTH)2	0.001 (0.029)	
factor(MONTH)3	0.0001 (0.027)	
factor(MONTH)4	0.058** (0.028)	

factor(MONTH)5	0.072** (0.028)	
factor(MONTH)6	0.101*** (0.028)	
factor(MONTH)7	0.177*** (0.027)	
factor(MONTH)8	0.193*** (0.027)	
factor(MONTH)9	0.160*** (0.028)	
factor(MONTH)10	0.101*** (0.028)	
factor(MONTH)11	0.014 (0.028)	
factor(MONTH)12	0.009 (0.028)	
UNEM	-0.015*** (0.006)	
FEB		0.035 (0.037)
MAR		0.042 (0.039)
APR		0.099** (0.038)
MAY		0.057 (0.037)
JUN		0.054 (0.035)
JUL		0.088*** (0.033)
AUG		0.059 (0.040)
SEP		0.007 (0.038)
OCT		-0.032 (0.035)
NOV		-0.059* (0.035)
DEC		0.027 (0.036)
SPDLAW	0.067*** (0.021)	-0.007 (0.024)
BELTLAW	-0.030 (0.023)	0.001 (0.027)
WKENDS	0.001 (0.006)	0.007 (0.007)
Constant	1.030*** (0.103)	-0.127 (0.105)
Observations	108	107
R ²	0.717	0.344
Adjusted R ²	0.668	0.227
Residual Std. Error	0.058 (df = 91)	0.067 (df = 90)

F Statistic 14.439*** (df = 16; 91) 2.945*** (df = 16; 90)

Note: $p < 0.1$; **$p < 0.05$** ; $p < 0.01$

There are very interesting results, since with the regression with differences we see that UNEM, SPDLAW and time trend variables became insignificant. In the question number #v) we have noticed counterintuitive results, and now we see that these results are not true, because variables are statistically insignificant, because we have removed the linear time trend and most of serial correlation. However, there is still some evidence for seasonality of fatal accidents.

#viii)

This is a very difficult question for even professional econometricians. We have rejected unit root for PRCFAT and accepted it for UNEM, but since we did the difference regression, we have noticed that other variables can not actually explain the changes in PRCFAT. Hence, the results from difference regression DO not always give results similar to levels, but we still need to do difference regression if we suspect a unit root.

#ix)

```
Fit9=lm(PRCFAT~t+factor(MONTH)+UNEM+SPDLAW+BELTLAW+WKENDS, data=Traffic)
```

```
U=Traffic$Fit_resid=Fit9$residuals
```

```
U_lag1=Traffic$Fit_resid_lag1=Lag(Traffic$Fit_resid, 1)
```

```
Fit_AR1=lm(U~U_lag1, data=Traffic)
```

```
summaryFit_AR1=summary(Fit_AR1)
```

```
summaryFit_AR1
```

Call: lm(formula = U ~ U_lag1, data = Traffic)

Residuals: Min 1Q Median 3Q Max -0.11681 -0.03358 0.00356 0.03112 0.11403

Coefficients: Estimate Std. Error t value Pr(>|t|)

(Intercept) 0.0002981 0.0049684 0.060 0.95226

U_lag1 0.2816414 0.0943463 2.985 0.00353 ** — Signif. codes: 0 '0.001' 0.01 '0.05' 0.1 '0.1' 1

Residual standard error: 0.05139 on 105 degrees of freedom (1 observation deleted due to missingness)

Multiple R-squared: 0.07823, Adjusted R-squared: 0.06945 F-statistic: 8.911 on 1 and 105 DF, p-value:

0.003526

The results of the regression show that the coefficient estimate of U_lag1 is 0.282 and highly significant with t-value of 2.985 and p-value of 0.00353. Hence, we can conclude that the errors AR(1) are serially correlated. The idea of strict exogeneity does not seem to make sense, since strictly exogenous explanatory variables can not react to what has happened to Yt in the past. Our sample contains only 108 variables, which I consider a small sample (small n), hence the strict exogeneity assumption does not make sense, in my opinion.

```
Fit_AR2=lm(U~U_lag1+t+factor(MONTH)+UNEM+SPDLAW+BELTLAW+WKENDS, data=Traffic)
```

```
summaryFit_AR2=summary(Fit_AR2)
```

```
summaryFit_AR2
```

Call: lm(formula = U ~ U_lag1 + t + factor(MONTH) + UNEM + SPDLAW + BELTLAW + WKENDS, data = Traffic)

Residuals: Min 1Q Median 3Q Max -0.11608 -0.03370 0.00380 0.03206 0.11424

Coefficients: Estimate Std. Error t value Pr(>|t|)

(Intercept) -7.837e-03 1.012e-01 -0.077 0.93842

U_lag1 2.840e-01 1.028e-01 2.762 0.00697 ** t -4.738e-06 4.198e-04 -0.011 0.99102

factor(MONTH)2 -2.295e-03 2.883e-02 -0.080 0.93674

factor(MONTH)3 -3.581e-03 2.747e-02 -0.130 0.89656

factor(MONTH)4 -3.447e-03 2.790e-02 -0.124 0.90196

factor(MONTH)5 -3.877e-03 2.786e-02 -0.139 0.88962

```
factor(MONTH)6 -2.980e-03 2.814e-02 -0.106 0.91589
factor(MONTH)7 -3.631e-03 2.725e-02 -0.133 0.89431
factor(MONTH)8 -3.789e-03 2.753e-02 -0.138 0.89085
factor(MONTH)9 -3.172e-03 2.825e-02 -0.112 0.91085
factor(MONTH)10 -3.856e-03 2.776e-02 -0.139 0.88981
factor(MONTH)11 -3.127e-03 2.812e-02 -0.111 0.91170
factor(MONTH)12 -3.625e-03 2.788e-02 -0.130 0.89685
UNEM -2.877e-04 5.457e-03 -0.053 0.95807
SPDLAW -2.054e-03 2.003e-02 -0.103 0.91858
BELTLAW 1.308e-03 2.255e-02 0.058 0.95389
WKENDS 1.044e-03 5.992e-03 0.174 0.86210
— Signif. codes: 0 ‘’ 0.001 ’’ 0.01 ’’ 0.05 ‘.’ 0.1 ’’ 1
```

Residual standard error: 0.05579 on 89 degrees of freedom (1 observation deleted due to missingness)
Multiple R-squared: 0.07908, Adjusted R-squared: -0.09682 F-statistic: 0.4496 on 17 and 89 DF, p-value: 0.9679

#This regression does not require strict exogeneity. We get the estimate of 0.284, with t-value of 2.762 and p-value of 0.00697, which is statistically significant. Hence, there is serial correlation of error terms.

```
#x)
vcovHAC=vcovHAC(Fit9, weights=bwNeweyWest)
vcovHAC1=vcovHAC(Fit9, weights=bwAndrews)
coeftest(Fit9, vcov.=vcovHAC)
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
--	----------	------------	---------	----------

```
(Intercept) 1.0298e+00 2.1202e-01 4.8571 4.932e-06 t -2.2352e-03 1.0120e-03 -2.2088 0.0297017
factor(MONTH)2 8.6077e-04 5.8550e-02 0.0147 0.9883025
factor(MONTH)3 9.2279e-05 5.8177e-02 0.0016 0.9987379
factor(MONTH)4 5.8220e-02 6.0229e-02 0.9667 0.3362805
factor(MONTH)5 7.1639e-02 6.2730e-02 1.1420 0.2564384
factor(MONTH)6 1.0126e-01 6.9915e-02 1.4484 0.1509537
factor(MONTH)7 1.7661e-01 7.7110e-02 2.2904 0.0243090 *
factor(MONTH)8 1.9261e-01 5.6564e-02 3.4052 0.0009855 factor(MONTH)9 1.6002e-01 6.4563e-02 2.4784
0.0150394
factor(MONTH)10 1.0104e-01 6.7305e-02 1.5012 0.1367759
factor(MONTH)11 1.3949e-02 6.5797e-02 0.2120 0.8325799
factor(MONTH)12 9.2005e-03 5.8020e-02 0.1586 0.8743557
UNEM -1.5426e-02 1.3354e-02 -1.1551 0.2510555
SPDLAW 6.7088e-02 4.3734e-02 1.5340 0.1285005
BELTLAW -2.9505e-02 5.2516e-02 -0.5618 0.5756133
WKENDS 6.2587e-04 1.2936e-02 0.0484 0.9615183
— Signif. codes: 0 ‘’ 0.001 ’’ 0.01 ’’ 0.05 ‘.’ 0.1 ’’ 1
```

```
coeftest(Fit9, vcov.=vcovHAC1)
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
--	----------	------------	---------	----------


```
(Intercept) 1.0298e+00 1.0065e-01 10.2318 < 2.2e-16 t -2.2352e-03 5.5458e-04 -4.0305 0.0001154
factor(MONTH)2 8.6077e-04 1.6181e-02 0.0532 0.9576911
factor(MONTH)3 9.2279e-05 2.3226e-02 0.0040 0.9968387
factor(MONTH)4 5.8220e-02 2.6738e-02 2.1774 0.0320341 *
factor(MONTH)5 7.1639e-02 2.9237e-02 2.4503 0.0161873 *
factor(MONTH)6 1.0126e-01 3.1119e-02 3.2540 0.0015980 ** factor(MONTH)7 1.7661e-01 3.4951e-02 5.0531
2.230e-06 factor(MONTH)8 1.9261e-01 2.4712e-02 7.7941 1.020e-11 factor(MONTH)9 1.6002e-01 2.9083e-
02 5.5021 3.424e-07 factor(MONTH)10 1.0104e-01 3.0700e-02 3.2911 0.0014213 factor(MONTH)11 1.3949e-
02 3.0641e-02 0.4552 0.6500175
factor(MONTH)12 9.2005e-03 2.8378e-02 0.3242 0.7465230
UNEM -1.5426e-02 7.0727e-03 -2.1811 0.0317566
SPDLAW 6.7088e-02 2.5188e-02 2.6635 0.0091455 ** BELTLAW -2.9505e-02 3.2342e-02 -0.9123 0.3640196
WKENDS 6.2587e-04 5.2035e-03 0.1203 0.9045275
— Signif. codes: 0 ‘0.001’ 0.01 ‘0.05’ 0.1 ‘.’ 1
```

#The test with Newey-West standard errors demonstrates that both estimates of SPDLAW and BELT LAW are not significantly different from zero. The Andrews test on the contrary shows significance of SPDLAW and BELTLAW.

summaryFit5

Call: lm(formula = PRCFAT ~ t + factor(MONTH) + UNEM + SPDLAW + BELTLAW + WKENDS, data = Traffic)

Residuals: Min 1Q Median 3Q Max -0.132085 -0.032573 0.000621 0.038226 0.132080

Coefficients: Estimate Std. Error t value Pr(>|t|)

```
(Intercept) 1.030e+00 1.030e-01 10.003 2.52e-16 t -2.235e-03 4.208e-04 -5.312 7.66e-07 factor(MONTH)2
8.608e-04 2.900e-02 0.030 0.976383
factor(MONTH)3 9.228e-05 2.741e-02 0.003 0.997321
factor(MONTH)4 5.822e-02 2.782e-02 2.093 0.039152 *
factor(MONTH)5 7.164e-02 2.764e-02 2.592 0.011129 *
factor(MONTH)6 1.013e-01 2.809e-02 3.604 0.000510 factor(MONTH)7 1.766e-01 2.726e-02 6.479 4.64e-09
factor(MONTH)8 1.926e-01 2.744e-02 7.018 3.91e-10 factor(MONTH)9 1.600e-01 2.820e-02 5.674 1.64e-07
factor(MONTH)10 1.010e-01 2.767e-02 3.651 0.000435 * factor(MONTH)11 1.395e-02 2.814e-02 0.496
0.621345
factor(MONTH)12 9.201e-03 2.786e-02 0.330 0.741959
UNEM -1.543e-02 5.544e-03 -2.782 0.006563 SPDLAW 6.709e-02 2.057e-02 3.262 0.001560 ** BELTLAW
-2.951e-02 2.323e-02 -1.270 0.207287
WKENDS 6.259e-04 6.162e-03 0.102 0.919328
— Signif. codes: 0 ‘0.001’ 0.01 ‘0.05’ 0.1 ‘.’ 1
```

Residual standard error: 0.05751 on 91 degrees of freedom Multiple R-squared: 0.7174, Adjusted R-squared: 0.6677 F-statistic: 14.44 on 16 and 91 DF, p-value: < 2.2e-16

#HAC standard errors affect the significance of two policy variables. From part v) SPDLAW variable was significant, while BELTLAW variable was insignificant. NEwey-West standard errors demonstrated the lack of significance, while Andrews standard erros demonstrate that both policy variables are significant.

Q3

```

#i)
#The  $H_0$  for AR(1) would be that  $b_1=0$ , and for AR(2)  $H_0$  would be  $b_1=b_2=0$ .
#ii)
#Yes, AssumptionTS#3 holds under each of null hypotheses, because null hypotheses state that
  stock returns are serially uncorrelated, so we can safely assume that they are weakly depend
  ent.
#iii)
NYSE=read.csv(file="NYSE.csv", header=TRUE, as.is=TRUE, sep = ",", na.strings = c("NA", ".",
""))
NYSE$RET_lag1=Lag(NYSE$RET, 1)
NYSE$RET_lag2=Lag(NYSE$RET, 2)
#iv)
Fit_AR3=lm(RET~RET_lag1, data=NYSE)
summaryFit_AR3=summary(Fit_AR3)
summaryFit_AR3

```

```

##
## Call:
## lm(formula = RET ~ RET_lag1, data = NYSE)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.261  -1.302   0.098   1.316   8.065
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.17963    0.08074   2.225  0.0264 *
## RET_lag1     0.05890    0.03802   1.549  0.1218
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.11 on 687 degrees of freedom
## (2 observations deleted due to missingness)
## Multiple R-squared:  0.003481, Adjusted R-squared:  0.00203
## F-statistic: 2.399 on 1 and 687 DF, p-value: 0.1218

```

The estimates tell us that there is positive effect of 0.0589 on returns from one week to next.

```

Fit_AR4=lm(RET~RET_lag1+RET_lag2, data=NYSE)
summaryFit_AR4=summary(Fit_AR4)
summaryFit_AR4

```

```
##
## Call:
## lm(formula = RET ~ RET_lag1 + RET_lag2, data = NYSE)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.2969  -1.3214   0.1099   1.3478   7.9832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.18575    0.08115   2.289  0.0224 *
## RET_lag1     0.06032    0.03818   1.580  0.1146
## RET_lag2    -0.03807    0.03814  -0.998  0.3185
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.112 on 685 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.004819, Adjusted R-squared:  0.001914
## F-statistic: 1.659 on 2 and 685 DF, p-value: 0.1912
```

The estimates of b1 tell us that there is positive effect of 0.06032 on returns from one week to next, but estimate of b2 has a negative effect of 0.03807. Hence, from one week to next there is a positive effect, but the week before past week has a negative effect on returns.

#v)

#AR(1) H0: b1=0. We would not reject the null, since t-value of b1 estimate is 1.549 and p-value is 0.1218.

#AR(2) H0:b1=b2=0. Assuming homoskedasticity, we can run the standard F-test to test the null hypothesis:

```
linearHypothesis(Fit_AR4, c("RET_lag1=0", "RET_lag2=0"), test = "F")
```

```
## Linear hypothesis test
##
## Hypothesis:
## RET_lag1 = 0
## RET_lag2 = 0
##
## Model 1: restricted model
## Model 2: RET ~ RET_lag1 + RET_lag2
##
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1     687 3069.4
## 2     685 3054.6  2    14.792 1.6586 0.1912
```

#With a p-value of 0.1912, we can not reject the null hypothesis.

#vi)

```
bptest(Fit_AR3, studentize = FALSE)
```

```
##
## Breusch-Pagan test
##
## data:  Fit_AR3
## BP = 95.217, df = 1, p-value < 2.2e-16
```

```
bptest(Fit_AR4, studentize = FALSE)
```

```
##
## Breusch-Pagan test
##
## data: Fit_AR4
## BP = 113.44, df = 2, p-value < 2.2e-16
```

#We find that there is no homoscedasticity in both AR(1) and AR(2) models. Hence, we need to use heteroskedasticity robust standard errors:

```
vcovHC=vcovHC(Fit_AR3, type = "HC0")
linearHypothesis(Fit_AR3, "RET_lag1=0", vcov. = vcovHC)
```

```
## Linear hypothesis test
##
## Hypothesis:
## RET_lag1 = 0
##
## Model 1: restricted model
## Model 2: RET ~ RET_lag1
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F Pr(>F)
## 1     688
## 2     687  1 0.7264 0.3943
```

#Even with robust SE's we can not reject the null hypothesis. Hence, we had valid inference in v).

```
vcovHC1=vcovHC(Fit_AR4, type = "HC0")
linearHypothesis(Fit_AR4, c("RET_lag1=0", "RET_lag2=0"), vcov. = vcovHC1, test="F")
```

```
## Linear hypothesis test
##
## Hypothesis:
## RET_lag1 = 0
## RET_lag2 = 0
##
## Model 1: restricted model
## Model 2: RET ~ RET_lag1 + RET_lag2
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F Pr(>F)
## 1     687
## 2     685  2 0.8711 0.419
```

#Same here. we do not reject H_0 and hence we had valid inferences in part v).